

ECE 251C – Fall 2025 – Parhi
Quiz 3

Quiz Rules: No notes, computers, calculators, or phones; have only a pen/pencil; work independently.

Please show your work and provide any justification of your answers so that you can receive partial credit.

On my honor, I pledge that I have neither given nor received unpermitted aid on this exam.

signature:

name:

(a) _____ / 8 points

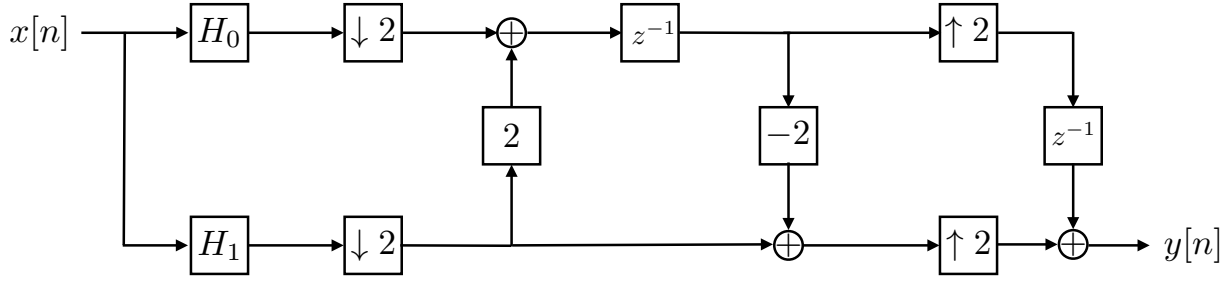
(b) _____ / 5 points

(c) _____ / 8 points

(d) _____ / 4 points

Total: _____ / 25 points

Consider the two-channel filter bank:



(a) Find the synthesis polyphase matrix $\mathbf{G}_p(z)$.

Solution.

$$\mathbf{G}_p(z) = \begin{bmatrix} z^{-1} & 2z^{-1} \\ -2z^{-1} & 1 - 4z^{-1} \end{bmatrix}$$

- If only $\mathbf{G}_p(z)$ was written and no work was shown, full points are awarded if the matrix is correct and no points are awarded if the matrix is incorrect.
- If some kind of factorization was written down and computed and the final matrix found was correct, then full points are awarded. If a small calculation error occurred, but a correct factorization was provided, then 6/8 points are awarded.
- If some kind of factorization was written down, but there were small errors in the factorization, then 4/8 points are awarded.
- If what was written does not make sense, then no points are awarded.

(b) Find the synthesis filters $G_0(z)$ and $G_1(z)$.

Solution. We have that

$$\begin{bmatrix} G_0(z) & G_1(z) \end{bmatrix} = \begin{bmatrix} z^{-1} & 1 \end{bmatrix} \mathbf{G}_p(z^2) = \begin{bmatrix} z^{-1} & 1 \end{bmatrix} \begin{bmatrix} z^{-2} & 2z^{-2} \\ -2z^{-2} & 1 - 4z^{-2} \end{bmatrix}$$

Therefore,

$$\begin{cases} G_0(z) = -2z^{-2} + z^{-3} \\ G_1(z) = 1 - 4z^{-2} + 2z^{-3} \end{cases}$$

- If the correct filters were determined that is consistent based on the answer to (a), then full points are awarded.
- If some kind of calculation went wrong, then 3/5 points are awarded.
- If what was written does not make sense, then no points are awarded.

(c) Find the PR analysis filters $H_0(z)$ and $H_1(z)$ by solving the equation $\mathbf{G}_p(z)\mathbf{H}_p(z) = z^{-K}\mathbf{I}$. What is the smallest K such that the analysis filters are causal?

Solution.

$$\mathbf{G}_p(z)^{-1} = \frac{1}{\Delta(z)} \begin{bmatrix} 1 - 4z^{-1} & -2z^{-1} \\ 2z^{-1} & z^{-1} \end{bmatrix},$$

where

$$\Delta(z) = z^{-1}(1 - 4z^{-1}) + 4z^{-2} = z^{-1}.$$

Thus, for causal solutions, we see that $\boxed{K=1}$ in which case

$$\mathbf{H}_p(z) = z^{-1}\mathbf{G}_p(z)^{-1} = \begin{bmatrix} 1 - 4z^{-1} & -2z^{-1} \\ 2z^{-1} & z^{-1} \end{bmatrix},$$

and so

$$\begin{bmatrix} H_0(z) \\ H_1(z) \end{bmatrix} = \mathbf{H}_p(z^2) \begin{bmatrix} 1 \\ z^{-1} \end{bmatrix} = \begin{bmatrix} 1 - 4z^{-2} - 2z^{-3} \\ 2z^{-2} + z^{-3} \end{bmatrix}$$

- If the correct answer is found via valid steps (and is consistent with the previous parts), then full points are awarded.
- If the inverse of $\mathbf{G}_p(z)$ is attempted to be computed, but something went wrong, then 4/8 points are awarded.
- 2 points are awarded for the correct determination of K .
- If what was written does not make sense, then no points are awarded.

(d) Determine L , where $y[n] = x[n - L]$, using your answer to (c).

Solution. $K = 1$ and so $L = 2K + 1 = 3$.

Either correct or incorrect. All or nothing, but if an incorrect K was determined in (c), but the correct answer of $2K + 1$ was provided, then full points are awarded.